ADVANCING GEOAI FOR DISASTER RESILIENCE: FROM DAMAGE ASSESSMENT TO INSIGHTFUL DECISION SUPPORT

A Dissertation Proposal

by

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Project Summary

Overview

Disaster resilience is a geographical question. It is rooted in space: resilience is place-based, shaped by local hazard exposure, social vulnerability, and human–environment interactions. It is diverse across the globe: disasters strike different places with different hazards and impacts, and the same type of damage looks different from one region to another. It is a geographic inquiry: beyond what happened, we ask where it happened, how severe it was, who was affected, and how to allocate limited response and recovery resources. And it is a cycle in time, from preparedness to response, recovery, and mitigation, in which each phase depends on timely, place-specific information. Disaster damage assessment is the hinge of this cycle: it tells responders where to go first, lets agencies verify eligibility and release aid, and gives planners a baseline for rebuilding. Imagery is its primary evidence. Overhead imagery from satellites and aircraft offers wide coverage and roof-level damage signals; ground-level street-view imagery captures the facades, debris, and flooding that people actually experience, but with sparse coverage. Geospatial Artificial Intelligence (GeoAI), the spatially explicit use of artificial intelligence on geographic data, is the engine that interprets these images at scale. Yet three challenges remain. Discriminative assessment is not yet accurate or reliable enough: models that see only one post-event view misclassify damage and stay confident when wrong. Specialized models do not generalize across events: each new disaster needs new labels and retraining, which is too slow for response. And assessment alone does not close the resilience loop: a damage label or map is not a decision, and each phase of the cycle needs actionable, accountable guidance.

This study traces and advances the evolution of GeoAI for disaster resilience along three evolving model families, discriminative, generative, and agentic, each answering one of the three challenges. Centered on seven interconnected case studies, the research: (1) begins with discriminative damage assessment to see better, restoring the pre-disaster baseline by pairing pre- and post-disaster street-view imagery and arbitrating among vision and vision-language models so that hyperlocal assessment becomes more accurate and more reliable; (2) advances to generative zero-shot understanding to respond faster, employing a reproducible multi-agent workflow that reasons across satellite and street-view evidence, regions, and hazards without event-specific training, synthesizing missing ground-level views from satellite imagery, and organizing the roles of generative AI across the disaster lifecycle; and (3) moves toward agentic decision support to prepare and deliver better, predicting street-level damage before an event from pre-event information and developing RAY (Resilience Assistant for You), an accountable GeoAI system that turns assessment into evidence-backed action for the public and decision-makers. Each stage builds upon the data, models, and lessons learned from the preceding one.

Intellectual Merit

This dissertation contributes new methods, models, and knowledge at each stage of disaster GeoAI. The discriminative stage develops a bi-temporal street-view damage classifier that uses the pre-disaster baseline as evidence and a multimodal arbitration framework that turns disagreement between vision and vision-language models into a reliability signal. The generative stage introduces RAPID, a reproducible zero-shot multi-agent workflow that transfers across hazards without fine-tuning; a satellite-conditioned model that synthesizes missing post-disaster street views; and a lifecycle framework mapping generative capabilities to disaster phases. The agentic stage produces a pre-event street-level damage prediction model and RAY, an accountable GeoAI system that links every claim to hashed evidence, checks outcome, process, and institutional validity, and abstains when unsupported.

Broader Impacts

This research benefits FEMA and local emergency managers, first responders, residents, insurers, urban planners, and the research community. For emergency operations centers, hyperlocal, reliability-aware damage maps can direct search, rescue, and debris operations to the hardest-hit

streets within hours. For residents and insurers, property-level evidence shortens the inspections that delay claims and rebuilding. For planners and under-resourced communities, zero-shot cross-view pipelines and pre-event damage prediction show likely impacts before a storm or fire, and RAY delivers this intelligence to non-experts as evidence-linked dossiers. For disaster scholars, the open datasets and code provide reusable multi-view benchmarks; for GeoAI and AI researchers, the reliability diagnostics and accountable-agent template transfer to other high-stakes uses of AI.

Project Description

1. Introduction and Significance

Disaster resilience is a geographical question for four reasons. It is rooted in space: resilience is place-based, shaped by local hazard exposure, social vulnerability, and human–environment interactions (Cutter et al., 2008). It is diverse across the globe: the 14 events in the BRIGHT building-damage dataset alone span five continents and seven disaster types, both natural and man-made, and the same type of building damage looks different in Morocco, Maui, and Noto, so methods that work in one place cannot be assumed to work everywhere (Chen et al., 2025). It is a geographic inquiry: beyond what happened, we ask where it happened, how severe it was, who was affected, and how to allocate limited resources, questions that volunteered geographic information answered faster than official channels during the 2009 Santa Barbara wildfires (Goodchild & Glennon, 2010). And it is a cycle in time: resilience is the ability to prepare for, absorb, recover from, and adapt to adverse events (National Academies, 2012), and each phase, from preparedness to response, recovery, and mitigation, depends on timely, place-specific information. Disaster damage assessment is the hinge of this cycle. In the immediate aftermath, it tells emergency responders where to go first and on what evidence; in the following weeks, it lets agencies verify eligibility and allocate limited personnel, supplies, and funds; and in the years that follow, site-specific assessment data underpin insurance claims, reconstruction, and resilience planning (Ghaffarian et al., 2018). FEMA's Damage Assessment Operations Manual formalizes the workflow: acquire imagery, classify damage severity, generate reports, and map impacts rapidly so that resources can be allocated (FEMA, 2016). The case studies in this dissertation target these same steps, and because two neighborhoods separated by a single block may experience vastly different levels of damage, they do so at the hyperlocal scale of streets and parcels.

Regarding disaster imagery, remote sensing technology has long provided an aerial perspective on disaster impacts. Satellite, aerial, drone, and LiDAR imagery have been widely used to map flood extents, roof damage, debris distribution, and burned structures following hurricanes, earthquakes, and wildfires (Hauptman et al., 2024; Kerle et al., 2020; Vetrivel et al., 2018); furthermore, the emergence of benchmark datasets such as xBD has established building damage classification based on nadir imagery as a standard deep learning task (Gupta et al., 2019). While this perspective offers broad coverage, it is limited to roofs and building footprints. Ground-level imagery (such as street-view images), by contrast, aligns more closely with human perceptions of disasters, capturing details like wall damage, fallen trees, and obstructed access points (Anguelov et al., 2010; Biljecki & Ito, 2021; Zhai & Peng, 2020). However, ground-level imagery has incomplete spatial coverage and often lacks temporal continuity, as obtaining pre- and post-disaster imagery for the same location is often difficult. Each perspective captures only one facet of the disaster's impact. Having this imagery, however, is not the same as understanding it, and three things break manual interpretation. The first is volume: satellites capture a whole disaster zone within days, yet only a small share of that imagery is ever analyzed. The second is speed: interpretation still depends on labels made by hand, one image pair at a time, and a single small town, Horseshoe Beach, Florida, required 2,556 labeled pairs. The third is complexity: the same building can look destroyed from the street and intact from above, so the views can disagree with each other. Manual interpretation cannot scale, which calls for AI that understands geography.

GeoAI, defined as spatially explicit artificial intelligence that leverages geographic data and spatial context to generate geographic information, knowledge, and services, is the engine that interprets such imagery at scale (Janowicz et al., 2020; Li & Hsu, 2022). The key word is spatially explicit: location, neighborhood, and scale are part of the model, not just metadata. Talking about GeoAI, however, requires being precise about what kind of AI is meant, and this dissertation organizes the models into three families with three roles. Discriminative models, task-specific classifiers, answer closed questions such as whether a building is damaged and whether the damage is mild, moderate, or severe; they are accurate on the task they were trained for, but they need labels for every event and stay confident even when wrong (Cheng et al., 2021; Guo et al., 2017; Nex et al., 2019). Generative models, including foundation and vision-language models, answer open questions such as why a location is severely damaged and what responders should do there; they generalize zero- and few-shot across hazards and regions, but they are usually less accurate than tuned classifiers and may hallucinate (Achiam et al., 2023; Bommasani et al., 2021; Mai et al., 2024; Radford et al., 2021). Agentic AI combines a large language model with tools and context: it can plan the analysis, call GIS tools and live data so that answers rest on computed evidence rather than memory, combine classifiers for accuracy with foundation models for reasoning, and extend beyond damage assessment to the whole resilience cycle (Li & Ning, 2023; Wang et al., 2024). This dissertation views these families as successive stages in the evolution of GeoAI for disaster resilience, a process through which disaster intelligence becomes more accurate and reliable, more transferable, and ultimately more accountable.

Each model family comes with a challenge, and viewed through this lens three challenges remain. First, discriminative assessment is not yet accurate or reliable enough. Most existing damage models see only one post-event view of a single event and discard the pre-disaster baseline, so they mistake pre-existing conditions for impacts and temporary debris for structural failure; and because fine-tuned networks are systematically overconfident, a confidently wrong severity label can misdirect a rescue team (Guo et al., 2017). Disagreement between views or between models is treated as noise rather than as information. The question is how to see better. Second, specialized models do not generalize across events. A classifier trained on one hurricane transfers poorly to another hazard or region, and each new disaster requires new labels, retraining, and validation, which is too slow for response; what is needed is assessment that transfers across hazards and regions without event-specific training and that explains its judgments rather than returning bare labels. The question is how to respond faster. Third, assessment alone does not close the resilience loop. A damage label or a map is not a decision: preparedness needs to anticipate impacts before an event, response and recovery need to inspect affected places afterward, and every phase needs guidance that is actionable and accountable. Current pipelines end with a label or a map, and the emerging generation of autonomous GeoAI agents that could deliver richer intelligence lacks mechanisms for evidence lineage, accountability, and refusal when evidence is insufficient. The question is how to prepare and deliver better.

This dissertation addresses these challenges through three objectives, each realized by a set of linked case studies that together follow GeoAI's evolution. Objective 1 addresses Challenge 1 (Cases 1–2): improve the accuracy and reliability of hyperlocal damage assessment by fusing pre- and post-event views and arbitrating among multiple models, so that we see better. Objective 2 addresses Challenge 2 (Cases 3–5): enable zero-shot assessment that transfers across hazards and regions without new labels or retraining, through a reproducible multi-agent workflow, synthesis of missing ground-level views, and a lifecycle framework, so that we respond faster. Objective 3 addresses Challenge 3 (Cases 6–7): build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters, through pre-event damage prediction and the RAY system, so that we prepare and deliver better. Each stage inherits the data, models, and lessons of the previous one, so that together these studies show how progressively more capable AI lets us see, describe, and govern disasters at the scale of streets and neighborhoods.

2. Literature Review and Conceptual Framework

2.1 Discriminative damage assessment: seeing better across views and times

Deep learning has transformed remote sensing for disaster response (Ma et al., 2019; Zhu et al., 2017). Convolutional networks and, more recently, vision transformers classify building damage from very-high-resolution satellite imagery (Gupta et al., 2019; Xu et al., 2019), segment flooded structures by fusing multi-resolution and multi-temporal sources (Rudner et al., 2019), and detect structural failures in oblique aerial and drone imagery (Kerle et al., 2020; Vetrivel et al., 2018). Post-hurricane aerial assessment has reached operational accuracy for coarse categories (Cheng et al., 2021). Nevertheless, overhead views cannot see façades, ground-floor flooding, or debris under tree canopy, and their damage classes are correspondingly coarse.

Street-view imagery partially fills this gap. Reviews document its rapid adoption in urban analytics and public health (Biljecki & Ito, 2021; Kang et al., 2020), and early disaster studies used Google Street View to document hurricane damage at the parcel level (Zhai & Peng, 2020). Multimodal models that combine street-view images with structured building data improve post-hurricane assessment (Zhang et al., 2026), and cross-view geolocalization now links ground photographs to satellite tiles for disaster mapping (Li et al., 2025). Yet almost all street-view damage models use only post-event images, so they mistake pre-existing conditions for impacts and temporary debris for structural failure. Moreover, accuracy-centered evaluation hides a second problem: modern neural networks are poorly calibrated, and their most confident predictions are not necessarily their most correct ones (Guo et al., 2017). In disaster response, an overconfident error is operationally worse than an uncertain one, because a low-confidence prediction can be flagged for human review while a confident one is acted upon. Contrastive vision-language models such as Contrastive Language-Image Pre-training (CLIP) reason about scenes through natural-language descriptions and exhibit a different error profile than purely visual classifiers (Radford et al., 2021), suggesting that combining, rather than replacing, model families can improve reliability. Interpretability tools such as Gradient-weighted Class Activation Mapping (Grad-CAM) can confirm whether a classifier attends to real damage evidence rather than background (Selvaraju et al., 2017), and interpretability is increasingly regarded as a prerequisite for deploying machine learning in consequential settings (Doshi-Velez & Kim, 2017). The discriminative stage of this dissertation therefore asks how baselines, model disagreement, and interpretability can make assessment from a single, incomplete ground-level view more accurate and more reliable.

2.2 Generative zero-shot understanding: responding faster across views, regions, and hazards

Large language models and multimodal foundation models open a further path. Foundation models transfer across tasks with few or no labels (Bommasani et al., 2021; Mai et al., 2024), and multimodal models can describe a damaged scene, weigh evidence from different views, and generate reports for human users (Achiam et al., 2023). Applied to disasters, this promises zero-shot assessment that does not require event-specific training and can extend from one hurricane to many events and hazard types. Generative vision models add a complementary capability: when post-event street-level coverage is missing altogether, cross-view image synthesis from satellite to ground, which has matured in computer vision through conditional generative adversarial networks, geometry-guided panorama synthesis, and diffusion models (Ho et al., 2020; Regmi & Borji, 2018; Rombach et al., 2022; Shi et al., 2022), could reconstruct a plausible ground-level view of the same place.

Three limitations remain. First, results are typically reported for one model on one dataset, so reproducibility across models, regions, and hazards is unknown. Second, generated reasoning is rarely checked for consistency with the underlying imagery, so fluent but unsupported explanations pass unnoticed; likewise, cross-view synthesis has not been adapted to represent disaster damage or evaluated for trustworthiness beyond pixel-level realism. Third, generative AI is usually studied for a single phase of the disaster cycle, leaving open how it should be organized across preparedness, response, recovery, and mitigation. This dissertation's generative stage responds to these limitations with a reproducible multi-agent workflow, a structure-aware evaluation of synthesized views, and a lifecycle framework.

2.3 Agentic decision support: preparing and delivering better with accountable agents

Place-based resilience frameworks emphasize that assessment should inform decisions before, during, and after events (Cutter et al., 2008). Two developments make this feasible for damage assessment. Earth-observation foundation models such as AlphaEarth now provide globally consistent 10-m embeddings that transfer to diverse mapping tasks with sparse labels (Brown et al., 2025), offering a scalable pre-event description of the physical environment that can be combined with hazard forecasts and street-level imagery. At the same time, autonomous GIS and GeoAI agents built on large language models can plan analyses, call spatial tools, and converse with users (Li & Ning, 2023), suggesting a delivery layer where residents and planners can query disaster intelligence directly. Autonomy, however, raises the accountability problems long recognized in AI safety: agents can assert claims without evidence, act outside their competence, and leave no trace of how they reached a conclusion (Amodei et al., 2016; Doshi-Velez & Kim, 2017). What is missing is a GeoAI architecture in which every spatial claim carries its evidence lineage, unknowns are declared rather than filled in, and the system refuses to answer when evidence is insufficient. The agentic stage of this dissertation supplies that architecture.

2.4 Conceptual Framework

Building on these gaps, this dissertation develops a conceptual framework in which GeoAI for disaster resilience evolves through three stages, each responding to one of the three challenges, each contributing one insight, and each building on the last (Figure 1). Stage 1, discriminative damage assessment, responds to the challenge that discriminative assessment is not yet accurate or reliable enough: pre-trained vision models restore the pre-disaster baseline by pairing pre- and post-disaster street-view imagery, and disagreement among vision and vision-language models is arbitrated rather than ignored, so that we see better; the insight is more accurate and reliable hyperlocal damage assessment (Cases 1–2). Stage 2, generative zero-shot understanding, responds to the challenge that specialized models do not generalize across events: structured zero-shot reasoning by multimodal agents replaces event-specific training, extends assessment from one view and one event to satellite and street-view evidence across regions and hazards, produces interpretable reports, synthesizes ground-level views where they are missing, and is situated within the full disaster lifecycle, so that we respond faster; the insight is rapid, interpretable, zero-shot assessment across views, regions, and hazards (Cases 3–5). Stage 3, agentic decision support, responds to the challenge that assessment alone does not close the resilience loop: damage is anticipated before events, affected places are inspected afterward, and place-based impacts are communicated through an accountable GeoAI system that ties every claim to its evidence and abstains when evidence is insufficient, so that we prepare and deliver better; the insight is preparedness and accountable place-based decision support (Cases 6–7).

The research plan consists of seven case studies organized into three components. The first component includes two discriminative studies: bi-temporal street-view damage assessment with pre-trained vision models, and a multimodal arbitration framework that improves reliability. The second component comprises three generative studies: RAPID, a reproducible multi-agent workflow for interpretable damage assessment from satellite and street-view imagery across hurricanes and wildfires; generative satellite-to-street synthesis of post-disaster views; and a lifecycle framework for generative AI in disaster resilience. The third component develops two agentic studies: a preparedness-oriented model that predicts and visualizes street-level impacts from pre-event information, and RAY (Resilience Assistant for You), an accountable GeoAI system for place-based disaster intelligence. Together, these studies move from classifying damage in one place and one view, to understanding damage across views, regions, and hazards, to anticipating, inspecting, and communicating impacts with accountable agents.

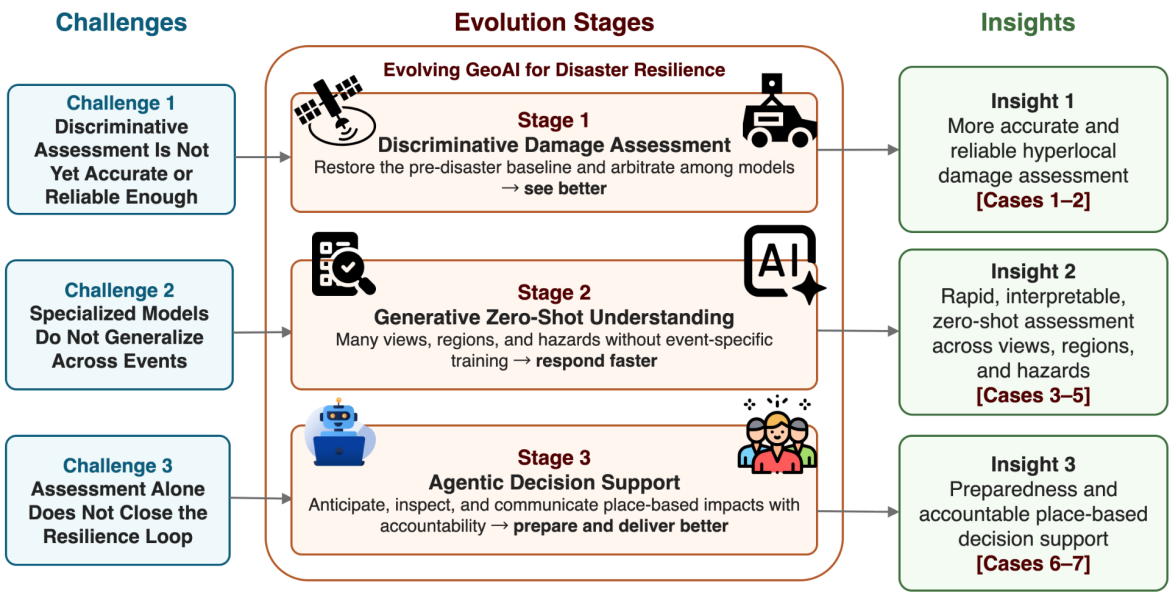


Figure 1. Conceptual framework connecting the three challenges to the three stages of evolving GeoAI for disaster resilience (discriminative, generative, and agentic) and the insight each stage contributes (Cases 1–7).

3. Research Questions and Hypotheses

Central Question. How does the evolution of GeoAI from discriminative to generative to agentic models make disaster damage assessment more accurate and reliable, more rapid and transferable across views, regions, and hazards, and more useful for preparedness and accountable decision support at the hyperlocal scale?

RQ1: Discriminative Damage Assessment: See Better (Objective 1, Case studies 1–2). Can fusing pre- and post-disaster street-view imagery and arbitrating among vision and vision-language models make hyperlocal damage assessment of a single hurricane more accurate and more reliable?

H1a. Pairing pre- and post-disaster street-view imagery will improve damage classification over post-event-only models and shift model attention toward true post-event changes.

H1b. Arbitrating between complementary model families (an image-only vision model and a vision-language model) will raise accuracy while reducing the share of overconfident errors relative to either model alone.

RQ2: Generative Zero-Shot Understanding: Respond Faster (Objective 2, Case studies 3–5). Can generative AI enable zero-shot assessment that transfers across views, regions, and hazards without new labels or retraining, through multi-agent reasoning, synthesis of missing ground-level views, and a lifecycle organization of generative capabilities?

H2a. Without event-specific training, structured multi-agent reasoning will achieve severity accuracy comparable to supervised baselines across hurricane and wildfire events, will produce reports whose stated evidence is consistent with the imagery, and will remain reproducible across model families.

H2b. Satellite-conditioned generative models will produce post-disaster street views whose structure and damage semantics are consistent with the overhead evidence.

H2c. The actionability of generated disaster intelligence will vary systematically by disaster phase.

RQ3: Agentic Decision Support: Prepare and Deliver Better (Objective 3, Case studies 6–7). To what extent can street-level damage be anticipated from information available before an event, and can an accountable agentic GeoAI system deliver place-based disaster intelligence to the public and decision-makers with evidence lineage and explicit refusal boundaries?

H3a. Pre-disaster street-view imagery, Earth-observation embeddings, hazard forecasts, and building context will predict post-event street-level severity well above chance under leakage-audited, street-segment-grouped validation and will add information beyond FEMA flood zones.

H3b. An agent architecture that ties every claim to hashed evidence and declared unknowns and checks outcome, process, and institutional validity will outperform an unconstrained agent on claim support, refusal correctness, and user trust without sacrificing usefulness.

4. Research Design and Methods

4.1 Overall Research Design

This dissertation comprises seven studies organized into three components that correspond to three generations of GeoAI. Studies 1 and 2 are discriminative: Study 1 pairs pre- and post-disaster street-view imagery of a single hurricane, and Study 2 arbitrates between an image-only and an image-text model to improve reliability. Studies 3, 4, and 5 are generative: Study 3 builds and validates RAPID, a zero-shot multi-agent workflow over satellite and street-view imagery across hurricane and wildfire events; Study 4 synthesizes post-disaster street views from satellite imagery where ground-level coverage is missing; and Study 5 situates such generative capabilities within a lifecycle framework and evaluates their actionability. Studies 6 and 7 are agentic: Study 6 predicts street-level damage from pre-event information, and Study 7 develops and evaluates RAY, an accountable agentic GeoAI system. Table 1 summarizes the mapping between components, case studies, and research questions.

Table 1. Research components, case studies, and research questions

Component	Case studies	Research question addressed
Discriminative damage assessment	Case studies 1–2: Bi-temporal street-view assessment and multimodal arbitration for reliability	RQ1 (see better): Can pre-disaster baselines and multimodal arbitration make hyperlocal damage assessment more accurate and more reliable?
Generative zero-shot understanding	Case studies 3–5: RAPID multi-agent pipeline, satellite-to-street synthesis, and generative AI across the disaster lifecycle	RQ2 (respond faster): Can generative AI enable zero-shot assessment across views, regions, and hazards without new labels or retraining?
Agentic decision support	Case studies 6–7: Pre-event street-level damage prediction and the RAY accountable GeoAI agent	RQ3 (prepare and deliver better): Can pre-event information anticipate street-level damage, and can an accountable agentic GeoAI system deliver place-based intelligence with evidence lineage?

4.2 Component 1: Discriminative damage assessment (Case studies 1–2)

4.2.1 Study Context and Data

Case Study 1 examines Horseshoe Beach, Florida, a coastal community severely affected by Hurricane Milton in October 2024. We collected post-disaster street-level imagery at the local road level with a vehicle-mounted Mosaic 51 camera system, and obtained pre-disaster imagery for the same road segments from Google Street View. Trained annotators geographically paired and labeled images as mild, moderate, or severe damage, yielding 2,556 pre/post pairs; the resulting Milton bi-temporal dataset and study-area boundary have been released publicly. This is, deliberately, the narrowest setting in the dissertation: one hurricane, one community, and one ground-level perspective

that is itself incomplete. Case Study 2 uses post-disaster images from the same dataset, along with natural-language scene descriptions generated by a large language model and, for comparison, written by human annotators.

4.2.2 Measurement and Analysis

Case Study 1 compares three model settings: post-disaster images only, pre-disaster images only as a control, and paired pre- and post-disaster images. Pre-trained Swin Transformer and ConvNeXt backbones (Liu et al., 2021; Liu et al., 2022) are fine-tuned under several bi-temporal designs, including six-channel stacking, cross-attention, Siamese feature difference, and dual-channel feature fusion, and are evaluated with accuracy and class-wise F1. Grad-CAM heatmaps (Selvaraju et al., 2017) test whether models attend to true pre/post changes rather than background. Completed results show that accuracy improved from 66.14% for the single-image Swin baseline to 77.11% for the dual-channel feature-fusion ConvNeXt, and that pre-disaster context concentrates attention on genuine changes. In other words, the pre-disaster baseline is not merely additional data; it shifts the task from recognizing damage to recognizing change.

Case Study 2 starts from the observation that using images alone is not enough, because an image-only model can be confidently wrong. A fine-tuned vision transformer may, for example, assign 70% to mild, 20% to moderate, and 10% to severe for a scene whose true label is severe. For every misclassified image we therefore compute a confidence gap C, the probability of the predicted class minus the probability assigned to the true class (here $0.70 - 0.10 = 0.60$), and sort errors into overconfident ($C \geq 0.40$), medium-confidence ($0.10 < C < 0.40$), and ambiguous ($C \leq 0.10$) errors. To add evidence beyond pixels, each image is also turned into text: a large language model describes the scene and names its damage cues, such as sagging power lines, scattered debris, or collapsed fragments, so that attention rests on the damage evidence rather than on the whole scene. An image-only vision transformer (Dosovitskiy et al., 2021), a text-only classifier, and a CLIP-based image-text model (Radford et al., 2021) are then compared not only on accuracy but on this confidence-stratified error profile. DamageArbiter accepts a prediction when the visual and image-text models agree and, when they disagree, arbitrates between them with a lightweight logistic-regression meta-classifier that uses only label-free confidence and uncertainty measures, so that it can be deployed on unlabeled imagery. We assess performance with accuracy, weighted F1, and Matthews correlation coefficient under out-of-fold cross-validation, and map results to compare predicted and ground-truth severity spatially, because reliability must be judged where errors occur, not only in aggregate. Completed results show that the image-only model was overconfident in about 71% of its errors whereas the image-text model produced no overconfident errors, and that arbitration raised accuracy to 75.8% with MCC of 0.62 while reducing the share of overconfident errors to about 16%.

4.3 Component 2: Generative zero-shot understanding (Case studies 3–5)

4.3.1 Study Context and Data

The generative stage widens the setting along every dimension that the discriminative stage held fixed. Case Study 3 begins with data preparation and assembles four datasets, about 1,100 images in total, that span views, hazards, and regions. Dataset A pairs post-disaster satellite and street-view imagery for Hurricane Ian in Florida, drawn from CVDisaster (Li et al., 2025). Dataset B is the Milton bi-temporal street-view set from Case Study 1, used both as post-event only and as pre/post pairs. Dataset C provides pre- and post-event satellite pairs with building-damage masks from BRIGHT (Chen et al., 2025). Dataset D contains post-disaster street-view images: geo-referenced wildfire images from the January 2025 Palisades and Eaton fires in Los Angeles County, matched to structure-level damage inspections published by CAL FIRE, and web-sourced multi-hazard scenes without geolocation. Together the datasets cover nine sites on four continents and earthquakes, floods, hurricanes, and wildfires, with ice storms and drought in the web-sourced subset. Case Study 4 draws on co-located satellite and street-view imagery from Hurricane Ian (2022) and Hurricane Milton (2024), aligned through geographic coordinates, to learn the mapping from an overhead view to a ground-level view of the same damaged place. Case Study 5 reuses these data to test generative models

across the disaster lifecycle and additionally draws on publicly available emergency-management documents to define preparedness and recovery tasks.

4.3.2 Measurement and Analysis

RAPID is organized in three stages. Stage 1 is data preparation: the remote sensing and street-view imagery described above is prepared and aligned around common locations. Stage 2 runs an autonomous multi-agent framework divided into specialized roles: a Disaster Perception Agent identifies hazard type and initializes the task; an Image Restoration Agent detects and repairs low-quality imagery using a quality score that combines contrast, sharpness, and a no-reference quality index and accepts a restoration only if quality improves; a Damage Recognition Agent identifies damaged objects and classifies severity from remote sensing and street-view inputs; and a Disaster Reasoning Agent analyzes impacts and generates a structured report. Stage 3 validates the models and generates structured disaster reports, which are checked by model-based consistency evaluation and by human review. All prompts are versioned and hashed so that runs can be reproduced, and the workflow is executed with multiple commercial and open multimodal models (including GPT-5-mini, ChatGPT-5.1, and the Gemini 2.5 and Gemini 3 families) to test model dependence. We evaluate severity predictions with accuracy and a normalized cross-severity error that penalizes distant misclassifications more heavily; we evaluate generated reports by checking each stated claim against the imagery for consistency, contradiction, or lack of support, and by a four-dimension rubric applied by both a language-model judge and human evaluators. Completed results are reported agent by agent. The perception agent recognizes six hazard types with an overall accuracy of 0.92 for the best model and F1-scores of at least 0.91 on drought, flood, hurricane, and wildfire for every model, with earthquake as the weak spot (recall of 0.50 to 0.58). For the remaining agents, the results show zero-shot severity accuracy between roughly 0.4 and 0.6 across models and events, with no single model winning everywhere, most errors falling between adjacent severity levels rather than between mild and severe, the strongest models approaching supervised baselines, restoration improving image quality on both views, and human and language-model judges broadly agreeing on the factuality, plausibility, completeness, and actionability of the generated reports. This work has been accepted at ACM SIGSPATIAL 2026.

Case Study 4 compares four generative vision models for synthesizing post-disaster street views from satellite imagery: a conditional generative adversarial network (Pix2Pix), Stable Diffusion with ControlNet conditioning based on latent diffusion (Rombach et al., 2022), a vision-language-model-guided variant, and a mixture of experts gated by severity. Two mechanisms inject disaster semantics: vision-language-model-guided prompts describing the overhead damage, and a mixture-of-experts decoder specialized by severity level. Because pixel metrics such as FID reward realism rather than correctness, a structure-aware evaluation framework compares the synthesized views with reference imagery on geometric consistency, semantic segmentation agreement, and severity classification by an independent damage model, with human raters judging plausibility. Completed results show that Pix2Pix stays pixel-aligned but blurry, ControlNet produces the most plausible scenes and the highest downstream severity consistency, vision-language-model-extracted semantics increase disaster realism, and the mixture of experts is most balanced across severity levels. Three lessons follow: single-model approaches cannot yet balance visual plausibility with strict structural alignment; high perceptual realism does not imply high semantic accuracy, so evaluation must be structure-aware; and trustworthiness, not fidelity alone, should be the target for disaster generation. This work has been accepted for presentation at the 2026 IEEE International Geoscience and Remote Sensing Symposium.

Case Study 5 develops a lifecycle framework that maps generative-AI capabilities, including description, restoration, synthesis, reasoning, and report generation, onto preparedness, response, recovery, and mitigation tasks, and evaluates a case study spanning cross-view and bi-temporal assessment. Beyond accuracy, domain evaluators score outputs for actionability on a five-point rubric, and the analysis identifies where generated intelligence is informative but not yet actionable. This gap between informative and actionable motivates the agentic stage: assessment alone does not close

the resilience loop. The chapter is under review for an edited volume on geography in the age of generative AI.

4.4 Component 3: Agentic decision support (Case studies 6–7)

4.4.1 Study Context and Data

Case Study 6 uses the Milton bi-temporal dataset as ground truth for post-event street-level severity and assembles the information that would have been available before landfall: pre-disaster street-view imagery, 64-dimensional AlphaEarth embeddings at 10-m resolution (Brown et al., 2025), National Hurricane Center wind and storm-surge forecasts issued before landfall, building footprints and parcel attributes, elevation, and American Community Survey estimates. Hurricane Ian (2022) in Lee County, Florida, provides a cross-event transfer test, and FEMA flood zones provide a regulatory benchmark for the resulting risk maps. Case Study 7 builds RAY on three tiers of data: a continuously updated watch of active United States hazards from USGS, National Weather Service, National Hurricane Center, and interagency wildfire feeds; deep case dossiers for Hurricane Milton and the Eaton Fire that integrate the damage products of Case Studies 1 to 6 with population, critical-facility, and social-vulnerability layers; and cross-view evidence from the assessment workflows. Every input is snapshotted with a cryptographic hash so that its provenance can be verified.

4.4.2 Measurement and Analysis

Case Study 6 frames preparedness as prediction, generation, and integration. Predict: a multimodal model encodes the pre-disaster street view, the AlphaEarth embedding of the surrounding cell, and tabular hazard and building features, and estimates severity probabilities and a continuous risk score for each street location; validation uses leakage-audited cross-validation grouped by street segment and a cross-event transfer test from Milton to Ian, so that performance reflects generalization to unseen neighborhoods rather than spatial autocorrelation. Generate: an instruction-tuned diffusion model trained on Milton pre/post image pairs, building on the generative model from Case Study 4, is conditioned on the predicted severity to produce a visual scenario of the expected damage, evaluated with the same structure-aware framework. Integrate: predictions and scenarios are combined into street-segment risk maps for inspection planning and benchmarked against FEMA flood zones, to test whether street-level prediction adds information beyond the regulatory hazard map.

Case Study 7 implements RAY (Resilience Assistant for You) as a static-first progressive web application that deploys large models within WebGIS and mobile interfaces, with an artifact-driven pipeline and a thin gateway around a large multimodal model. The system has a data plane, an agent plane, and a presentation plane, wrapped by a Steward Harness that checks four conditions on every product: outcome validity (is the claim supported by registered evidence), process validity (was it produced by an approved, reproducible workflow), institutional validity (does it respect the policies and mandates of the issuing agencies), and verifiability (can a user trace it to its sources). The harness enforces a policy of default deny: the agent may assert a claim about a place only if the claim cites registered evidence, unknowns are declared rather than inferred, planning trade-offs are made explicit, and answers that would require unavailable evidence are refused with an explanation. Dissemination is differentiated by tier: a resident mode returns an address-level resilience dossier with citation click-through, and a planner mode exposes trade-off sliders, a lineage viewer, and validity badges that report which checks each product passed. Evaluation combines a verifiable test environment of adversarial prompts and a policy matrix, measures of claim support and refusal correctness, and a user study with responders, planners, and residents comparing RAY with an unconstrained agent on trust, task completion, and comprehension.

5. Intellectual Merit

This dissertation contributes new methods, models, and knowledge to GeoAI for disaster resilience across the discriminative-to-generative-to-agentic trajectory.

(1) **In the discriminative stage, it reconceptualizes damage assessment as a change-detection and reliability problem rather than a single-image classification problem**. It contributes

a bi-temporal street-view damage classifier and a multimodal arbitration framework built on this view: no single post-event view sees a damaged place completely, so the pre-disaster baseline is restored as evidence of change, and disagreement between complementary models is treated as a reliability signal rather than as noise to be discarded. This perspective explains why accuracy and reliability must be measured together, and why an overconfident error is operationally worse than an uncertain one.

(2) **In the generative stage, it redefines what counts as a good assessment model when disasters are time-sensitive and place-specific**. It contributes RAPID, a reproducible zero-shot multi-agent workflow; a satellite-to-street synthesis model with a structure-aware evaluation; and a lifecycle framework for generative AI in disasters. By dividing assessment into specialized perception, restoration, recognition, and reasoning roles; replacing event-specific fine-tuning with reproducible zero-shot reasoning that produces inspectable reports; and synthesizing missing views, the research shifts evaluation from accuracy on one event to reliability, interpretability, and transferability across views, regions, and hazards.

(3) **In the agentic stage, it establishes accountability as a core design principle of GeoAI systems that close the resilience loop**. It contributes a pre-event street-level damage prediction model and RAY, an accountable agentic GeoAI system. Rather than treating autonomy and trust as opposites, the research formalizes how spatial judgments should link to verifiable hashed evidence, declared unknowns, and explicit refusal boundaries, checked for outcome, process, and institutional validity and for verifiability, and shows that place-based intelligence can be delivered to non-experts while remaining auditable. Human decision makers therefore remain the governors of disaster intelligence rather than its passive recipients.

6. Broader Impacts

The broader impacts of this dissertation reach four groups: emergency managers and first responders at FEMA and state and local agencies; residents and insurers; urban planners and community organizations; and the disaster and AI research communities.

(1) **It supports faster and more equitable emergency response and recovery**. For FEMA and local emergency operations centers, hyperlocal, reliability-aware damage maps can direct search, rescue, and debris operations to the hardest-hit streets within hours, when field inspection is slowest and most dangerous. For residents and insurers, property-level evidence can shorten the inspections that delay insurance payouts and rebuilding, with the largest benefits for communities that lack the resources for rapid manual assessment.

(2) **It advances the responsible use of generative AI for disasters**. Zero-shot cross-view workflows lower the barrier for agencies and planners without event-specific training data or specialist analysts, while the reliability diagnostics, consistency checks, and accountability mechanisms developed here provide safeguards that transfer to other consequential applications of foundation models in public health, climate adaptation, and urban planning.

(3) **It broadens public access to place-based disaster intelligence**. Pre-event damage prediction gives residents and planners a concrete picture of possible impacts before a storm or fire, and RAY delivers dossiers and planning trade-offs with evidence attached. Residents receive site-specific answers rather than a map they must interpret alone.

(4) **It benefits the research community**. For disaster scholars, the openly released multi-view datasets, prompts, code, and design documents provide reusable benchmarks for cross-view and cross-hazard damage assessment. For GeoAI and AI researchers, the reliability diagnostics, consistency checks, and accountable-agent template offer an evaluation protocol and a design pattern for autonomous GeoAI that other systems can adopt.

7. Current Status and Timeline

The dissertation is well advanced. Case studies 1 through 5 have been completed. Case study 1 was published in *Computers, Environment and Urban Systems* and was featured by Mosaic as an example of street-level damage assessment after Hurricane Milton; Case study 2 received a Best

Student Paper award at the 2025 International Cartographic Conference, won 2nd Place in the 2026 AAG GIS Specialty Group Student Honors Paper Competition, and its extended version is under review at *Natural Hazards*; Case study 3 has been accepted at ACM SIGSPATIAL 2026, with a companion abstract accepted for the CaGIS 2026 student paper competition; Case study 4 has been accepted at IEEE IGARSS 2026; and Case study 5 is under review as a chapter for the CRC Press volume *Geography in the Age of Generative AI*. Together, these studies complete the discriminative and generative stages, establish the bi-temporal, arbitration, multi-agent, and synthesis workflows on which the agentic stage builds, and have released the Milton bi-temporal dataset, prompts, and workflow code publicly. As of November 2026, the Ph.D. has yielded 8 first-authored publications (2 journal articles, 5 peer-reviewed conference papers, and 1 chapter under review), 14 co-authored publications, and 15 honors and awards.

Case study 7 is approximately 40% complete. The RAY prototype, including the live hazard watch, the Milton and Eaton deep-case dossiers, the Steward Harness with its policy, claim, and distribution rules, and the resident and planner interfaces, has been implemented, deployed, and covered by an automated evaluation environment of more than 350 tests; a short paper and demonstration have been submitted. Remaining work includes the user study and integration of the cross-view evidence tier. Case study 6 is about 20% complete: the pre-event data have been assembled, and AlphaEarth embeddings have been extracted for the study areas; model training, spatially blocked validation, and scenario visualization remain.

As shown in Figure 2, the preliminary examination was completed in Spring 2026, and the proposal defense takes place in November 2026. Case studies 6 and 7 will be completed between Fall 2026 and Spring 2027, with manuscripts submitted as each concludes. Dissertation writing has already begun and will proceed alongside the remaining analyses through Spring 2027, so that the complete dissertation can be circulated to the committee, revised, and defended, with the final defense and graduation planned for Summer 2027.

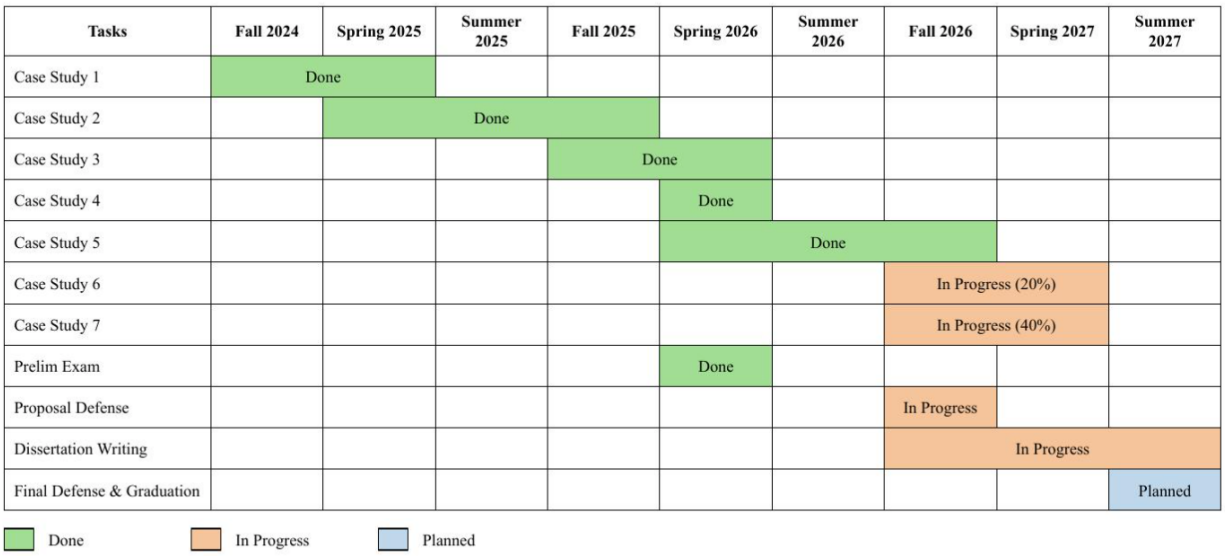


Figure 2. Timeline and current completion status of the seven case studies and major doctoral milestones.

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